



Sandhya M 250701639-2025-CSE ▾

S2

|              |                                |
|--------------|--------------------------------|
| Started on   | Sunday, 9 August 2026, 7:24 PM |
| State        | Finished                       |
| Completed on | Sunday, 9 August 2026, 7:32 PM |
| Time taken   | 8 mins 46 secs                 |
| Marks        | 1.00/1.00                      |
| Grade        | 10.00 out of 10.00 (100%)      |

**Question 1** | Correct | Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
```

```
{
```

```
    int i= 1;
```

```
    int s =1;
```

```
    while(s <= n)
```

```
    {
```

```
        i++;
```

```
        s += i;
```

```
    }
```

```
}
```

**Note:** No need of counter increment for declarations and scanf() and count variable printf() statements.

**Input:**

A positive Integer n

**Output:**

Print the value of the counter variable

**For example:**

| Input | Result |
|-------|--------|
| 9     | 12     |

**Answer:** (penalty regime: 0 %)

Ace editor not ready. Perhaps reload page?

Falling back to raw text area.

```
int main()
```

```
{
```

```
    int n;
```

```
    int i = 1;
```

```
    int s = 1;
```

```
    int count = 3;
```

```
    scanf("%d", &n);
```

```
    while (s <= n)
```

```
    {
```

```
        i++;
```

```
        count++;
```

```
        s += i;
```

```
        count++;
```

```
        count++;
```

```
    }
```

|   | Input | Expected | Got |   |
|---|-------|----------|-----|---|
| ✓ | 9     | 12       | 12  | ✓ |
| ✓ | 4     | 9        | 9   | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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S2

|              |                                   |
|--------------|-----------------------------------|
| Started on   | Sunday, 9 August 2026, 7:33 PM    |
| State        | Finished                          |
| Completed on | Tuesday, 11 August 2026, 10:44 AM |
| Time taken   | 1 day 15 hours                    |
| Marks        | 1.00/1.00                         |
| Grade        | 10.00 out of 10.00 (100%)         |

**Question 1** | Correct | Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
        printf("*");
    }
    else
    {
        for(int i=1; i<=n; i++)
        {
            for(int j=1; j<=n; j++)
            {
                printf("*");
                printf("*");
                break;
            }
        }
    }
}
```

**Note:** No need of counter increment for declarations and scanf() and count variable printf() statements.

**Input:**

A positive Integer n

**Output:**

Print the value of the counter variable

**Answer:** (penalty regime: 0 %)

```
1  #include <stdio.h>
2
3  int main()
4  {
5      int n;
6      int count = 0;
7
8      scanf("%d", &n);
9      count++;
10
11     if (n == 1)
12     {
13         count++;
14         printf("%d", count);
15     }
16     else
17     {
18         for (int i = 1; i <= n; i++)
19         {
20             count++;
21             for (int j = 1; j <= n; j++)
22             {
23                 count++;
24                 count++;
25                 count++;
26                 break;
27             }
28             count++;
29         }
30     }
31     count++;
32     printf("%d", count);
33 }
34 return 0;
35 }
```

|   | Input | Expected | Got  |   |
|---|-------|----------|------|---|
| ✓ | 2     | 12       | 12   | ✓ |
| ✓ | 1000  | 5002     | 5002 | ✓ |
| ✓ | 143   | 717      | 717  | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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S2

|              |                                   |
|--------------|-----------------------------------|
| Started on   | Tuesday, 11 August 2026, 10:46 AM |
| State        | Finished                          |
| Completed on | Tuesday, 11 August 2026, 10:49 AM |
| Time taken   | 2 mins 44 secs                    |
| Marks        | 1.00/1.00                         |

Grade 10.00 out of 10.00 (100%)

**Question 1** | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
    for (i = 1; i <= num; ++i)
    {
        if (num % i == 0)
        {
            printf("%d ", i);
        }
    }
}
```

**Note:** No need of counter increment for declarations and scanf() and counter variable printf() statement.

**Input:**

A positive Integer n

**Output:**

Print the value of the counter variable

**Answer:**

```
1  #include<stdio.h>
2
3  int main()
4  {
5      int num;
6      int i;
7      int count=0;
8
9      scanf("%d",&num);
10
11     for(i=1;i<=num;i++)
12     {
13         count++;
14
15         if(num%i ==0)
16         {
17             count++;
18         }
19         count++;
20     }
21     count++;
22
23     printf("%d", count);
24
25     return 0;
26 }
```

|   | Input | Expected | Got |   |
|---|-------|----------|-----|---|
| ✓ | 12    | 31       | 31  | ✓ |
| ✓ | 25    | 54       | 54  | ✓ |
| ✓ | 4     | 12       | 12  | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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S2

|              |                                   |
|--------------|-----------------------------------|
| Started on   | Tuesday, 11 August 2026, 10:49 AM |
| State        | Finished                          |
| Completed on | Friday, 14 August 2026, 9:19 AM   |
| Time taken   | 2 days 22 hours                   |
| Marks        | 1.00/1.00                         |
| Grade        | 10.00 out of 10.00 (100%)         |

**Question 1** | Correct | Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c = 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

**Note:** No need of counter increment for declarations and scanf() and count variable printf() statements.

**Input:**

A positive Integer n

**Output:**

Print the value of the counter variable

**Answer:**

```
1  #include <stdio.h>
2
3  int main()
4  {
5      int n, c = 0;
6      int count=1;
7
8      scanf("%d", &n);
9
10     count++;
11
12     for(int i = n/2; i < n; i++)
13     {
14         count++;
15
16         for(int j = 1; j < n; j = 2*j)
17         {
18             count++;
19
20             for(int k = 1; k < n; k = 2*k)
21             {
22                 count++;
23                 c++;
24                 count++;
25             }
26             count++;
27         }
28         count++;
29     }
30
31     printf("%d", count);
32
33     return 0;
34 }
```

|   | Input | Expected | Got |   |
|---|-------|----------|-----|---|
| ✓ | 4     | 30       | 30  | ✓ |
| ✓ | 10    | 212      | 212 | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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**Sandhya M 250701639-2025-CSE** ▾**S2****Started on** Tuesday, 11 August 2026, 10:54 AM**State** Finished**Completed on** Tuesday, 11 August 2026, 10:59 AM**Time taken** 4 mins 22 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

**Question 1** | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
    int rev = 0, remainder;
    while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n /= 10;
    }
    print(rev);
}
```

**Note:** No need of counter increment for declarations and scanf() and count variable printf() statements.

**Input:**

A positive Integer n

**Output:**

Print the value of the counter variable

**Answer:**

```
1  #include <stdio.h>
2
3  int main()
4  {
5      int n, rev = 0, remainder;
6      int count = 0;
7
8      scanf("%d", &n);
9
10     while (n != 0)
11     {
12         count++;
13         remainder = n % 10;
14         count++;
15         rev = rev * 10 + remainder;
16         count++;
17         n /= 10;
18         count++;
19     }
20
21     count++;
22     count++;
23     count++;
24
25     printf("%d", count);
26
27     return 0;
28 }
```

|   | Input | Expected | Got |   |
|---|-------|----------|-----|---|
| ✓ | 12    | 11       | 11  | ✓ |
| ✓ | 1234  | 19       | 19  | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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